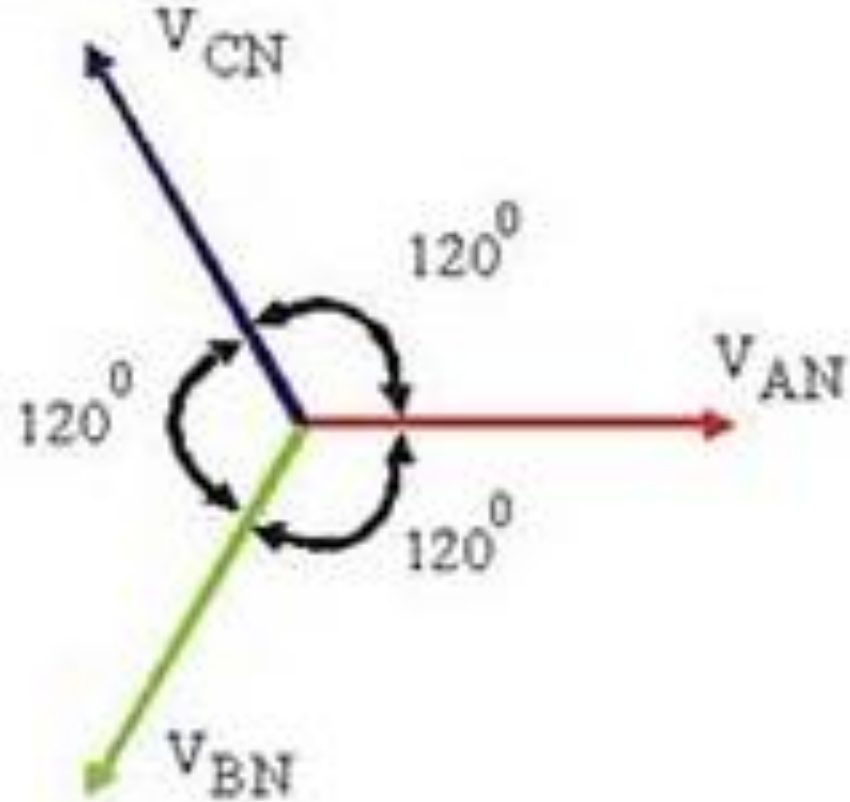
Three-Phase Uncontrolled Rectifiers

Chapter 4

Three-Phase Half-Wave Rectifiers

- Three-phase half-wave rectifiers are utilized in power electronics applications where the conversion of three-phase AC power to DC power is required. They are more efficient than their single-phase counterparts as they make use of all three phases of the AC power. However, the output DC voltage still experiences pulsations and needs additional filtering to achieve a stable DC voltage.



Vector diagram of 3-phase supply voltages

$$V_m = The\ Max\ of\ phase\ voltage$$

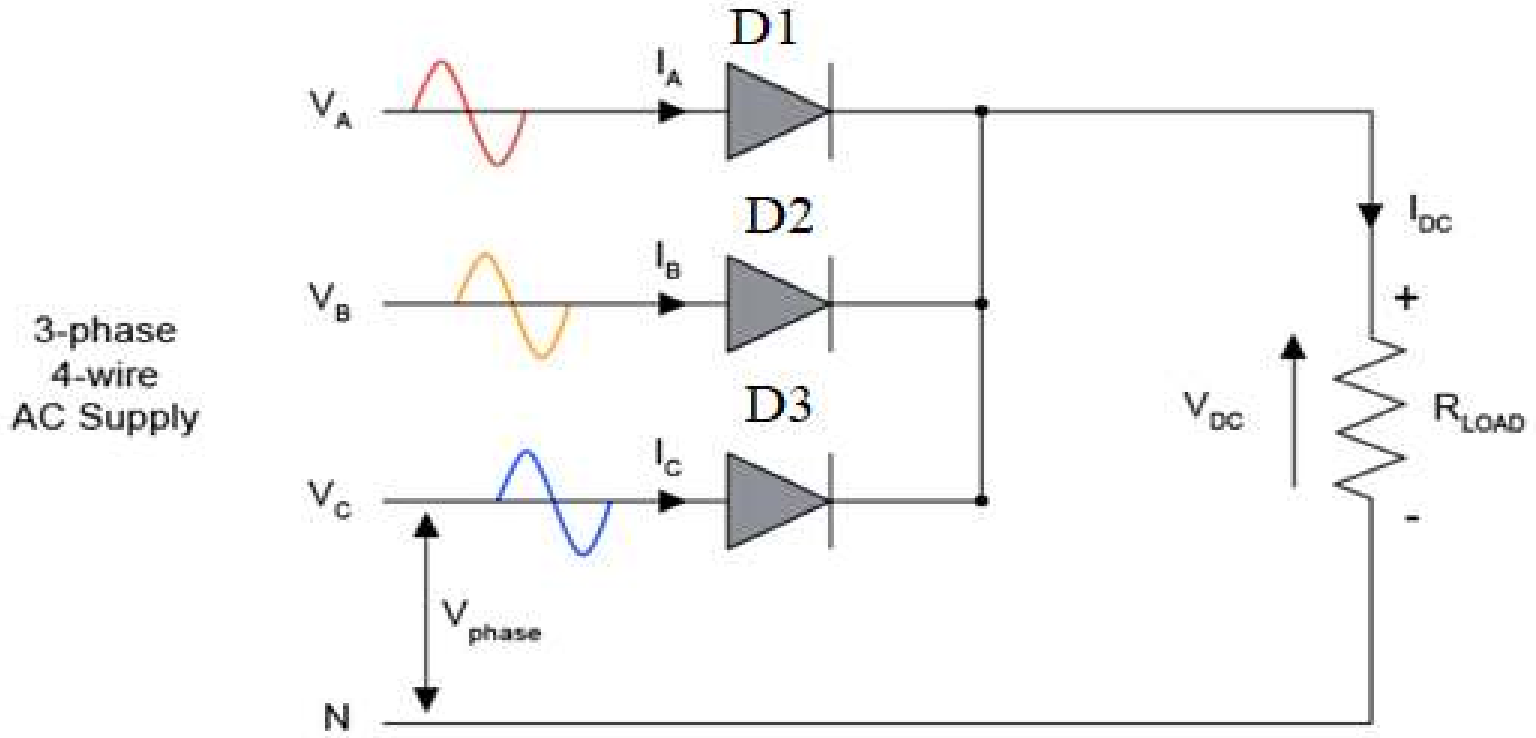
$$V_{an} = V_m \sin \omega t$$

$$V_{bn} = V_m \sin (\omega t - 120°)$$

$$V_{cn} = V_m \sin (\omega t + 120°)$$
$$V_{cn} = V_m \sin (\omega t - 240°)$$

Three Phase Half-Wave Rectifier with Resistive Load

The three-phase half-wave rectifier has the advantage of simplicity since it requires only three diodes. However, its output voltage is lower than other types of rectifiers since only one of the three phases is used at a time.



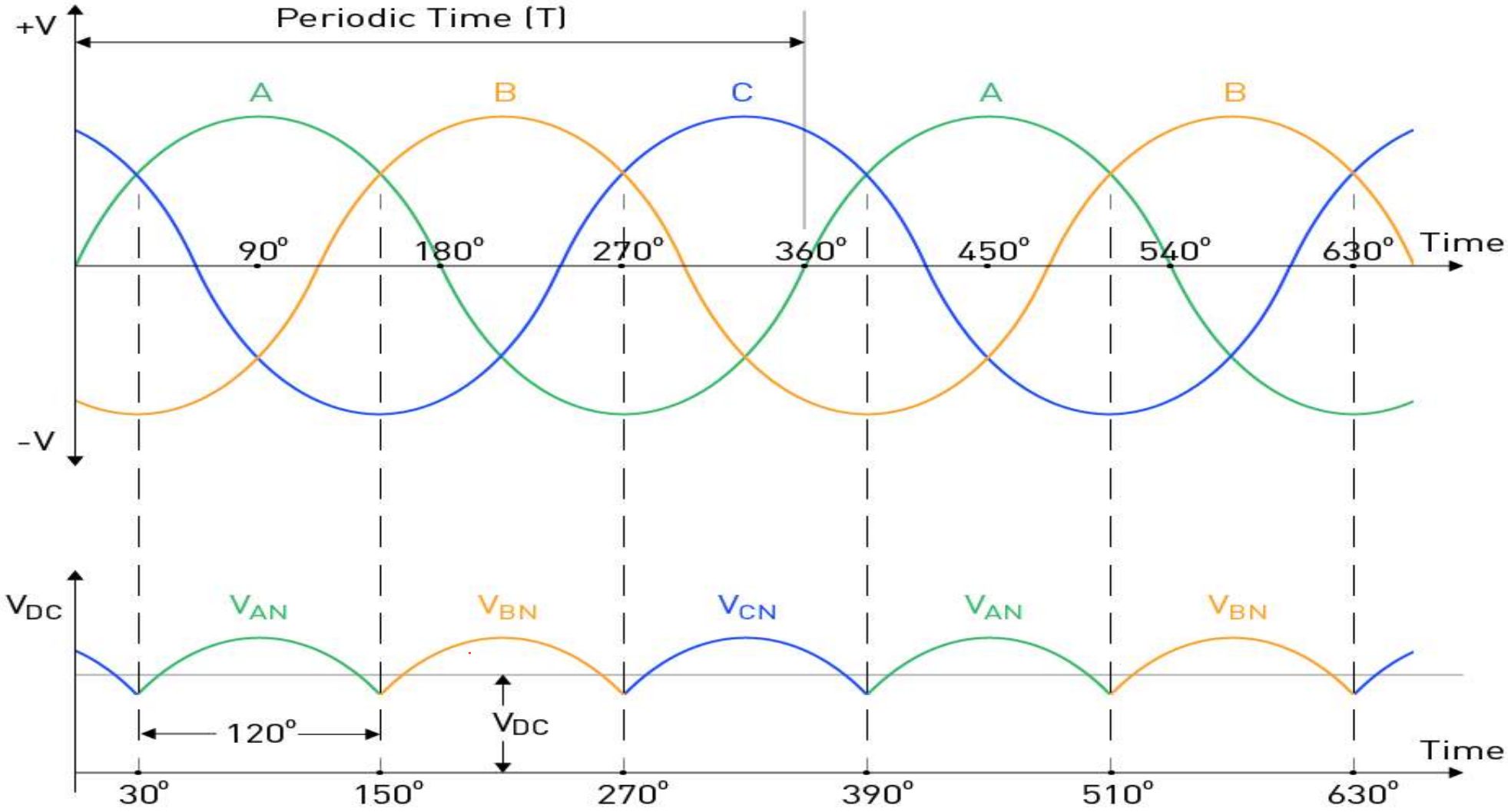
Let's assume that phase voltage V_A starts at 0°. During the time interval between 0° and 30° the voltage at the diode D_3 anode is the highest one ($VC > VA > VB$). Hence, diode D_3 conducts, and diodes D_1 and D_2 don't. Thereafter, during the time interval between 30° and 150° the voltage at the diode D_1 anode is the highest one ($30° - 90°$: $VA > VC > VB$; $90° - 150°$: $VA > VB > VC$).

Hence, diode D_1 conducts, and diodes D_2 and D_3 don't.

And so on… The waveform of the voltage at the output of the rectifier (on the load) is depicted in the figure below.

The time interval for the diodes conducting:

Period	On diode	Off diode	Diode Voltage		
			V_{D1}	V_{D2}	V_{D3}
$0-30^o$	D_3	D_1 and D_2	V_{AC}	V_{BC}	0
$30-150^o$	D_1	D_2 and D_3	0	V_{BA}	V_{CA}
$150-270^o$	D_2	D_3 and D_1	V_{AB}	0	V_{CB}
$270-390^o$	D_3	D_1 and D_2	V_{AC}	V_{BC}	0



Output Voltage Waveform

If the peak input voltage $V_{Phase\ Max}$, the average value of the output voltage waveform is:

$$V_{avg} = \frac{3}{2\pi} \int_{\pi/6}^{5\pi/6} V_{p,m}\ \sin(\omega t)\, d(\omega t)$$

$$V_{avg} = \frac{3\sqrt{3}}{2\pi} V_{phaseMax}$$

The *rms* load voltage is:

$$V_{rms} = \sqrt{\frac{3}{2\pi}\int_{\pi/6}^{5\pi/6} (V_{Pm})^2 \ \sin^2(\omega t)\, d\omega t}$$

$$= V_{Pm}\left[\frac{3}{2\pi}\left(\frac{\pi}{3} + \frac{1}{2}\sin\frac{2\pi}{3}\right)\right] = V_{Pm}\left[\frac{3}{2\pi}\left(\frac{\pi}{3} + \frac{\sqrt{3}}{4}\right)\right]$$

The load voltage form factor is:

$$FF = \frac{V_{rms}}{V_{avg}}$$

The ripple factor is:

$$\sqrt{(FF)^2 - 1}$$

The average diode current is:

$$I_{Davg} = \frac{I_{avg}}{3}$$

The rms diode current is:

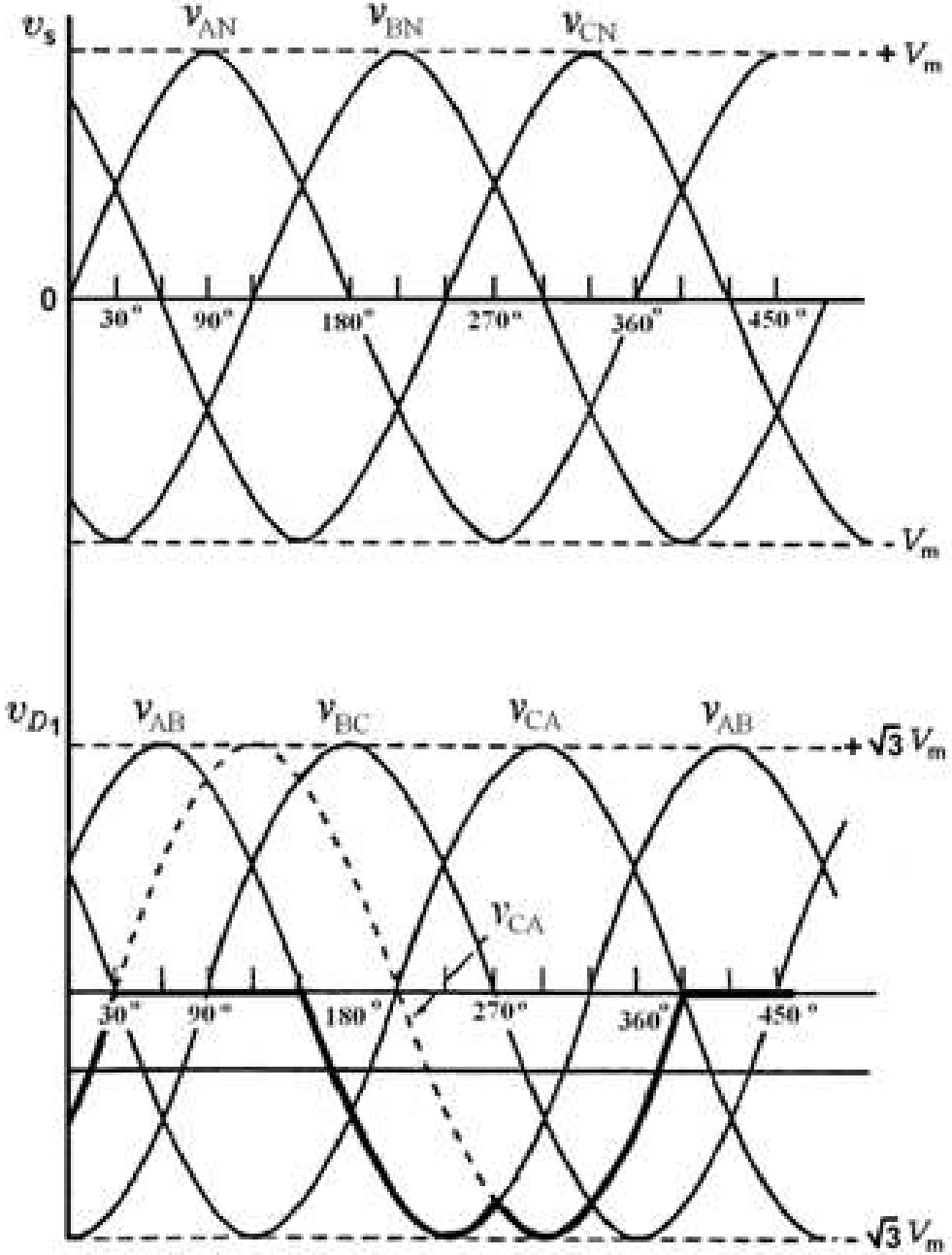
$$I_{Drms} = \frac{I_{rms}}{\sqrt{3}}$$

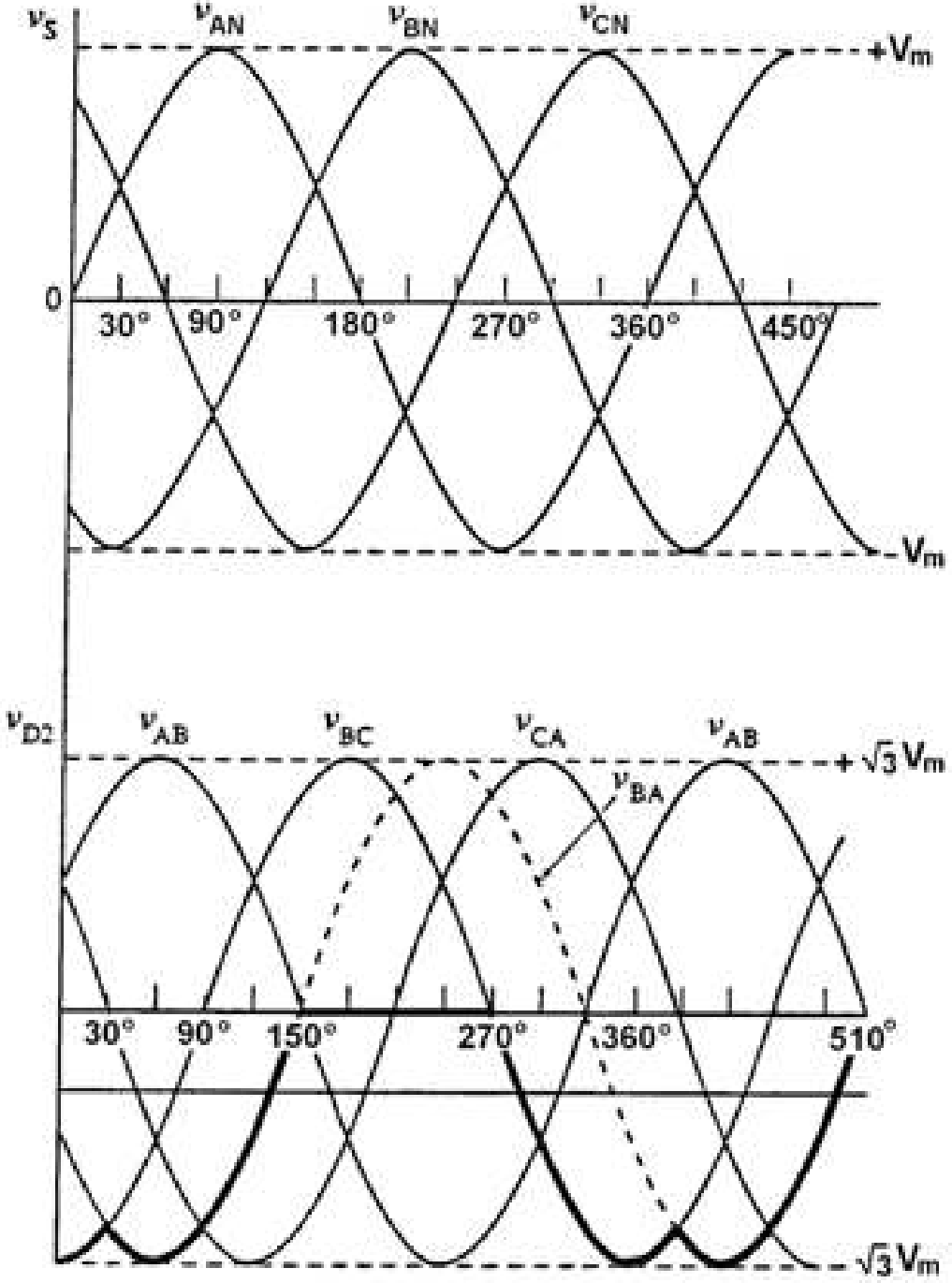
The diode conduction angle is:

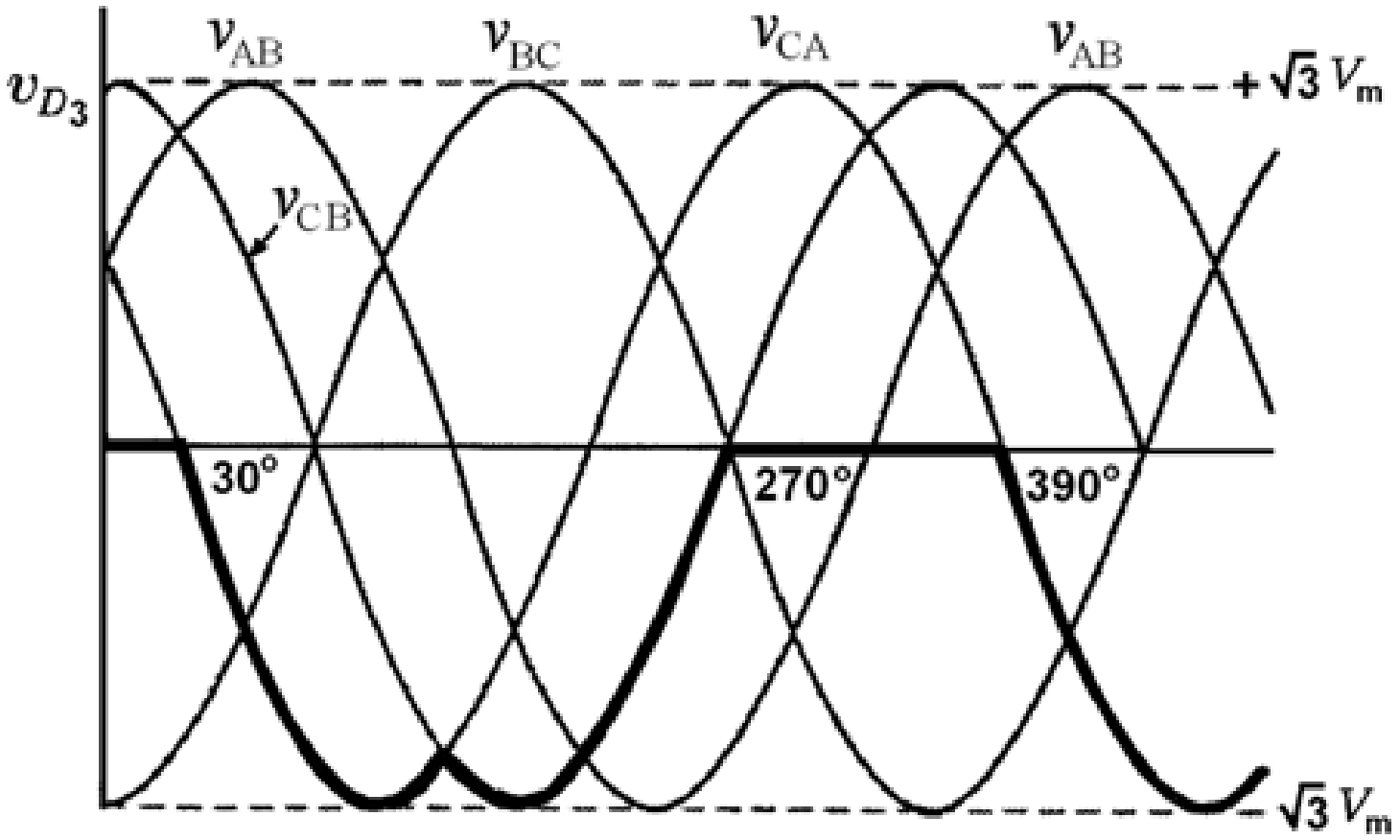
$$2\pi/3 \quad = 120^{\circ}$$

The peak diode reverse voltage is given by the maximum voltage between any two phases:

$$PIV = \sqrt{3}\ V_{Pm}. \quad = V_{L-L}$$


v_s
v_{AN}
v_{BN}
v_{CN}
$+V_m$
0
30°
90°
180°
270°
360°
450°
$-V_m$
v_{D1}
v_{AB}
v_{BC}
v_{CA}
v_{AB}
$+\sqrt{3}V_m$
v_{CA}
30°
90°
180°
270°
360°
450°
$-\sqrt{3}V_m$

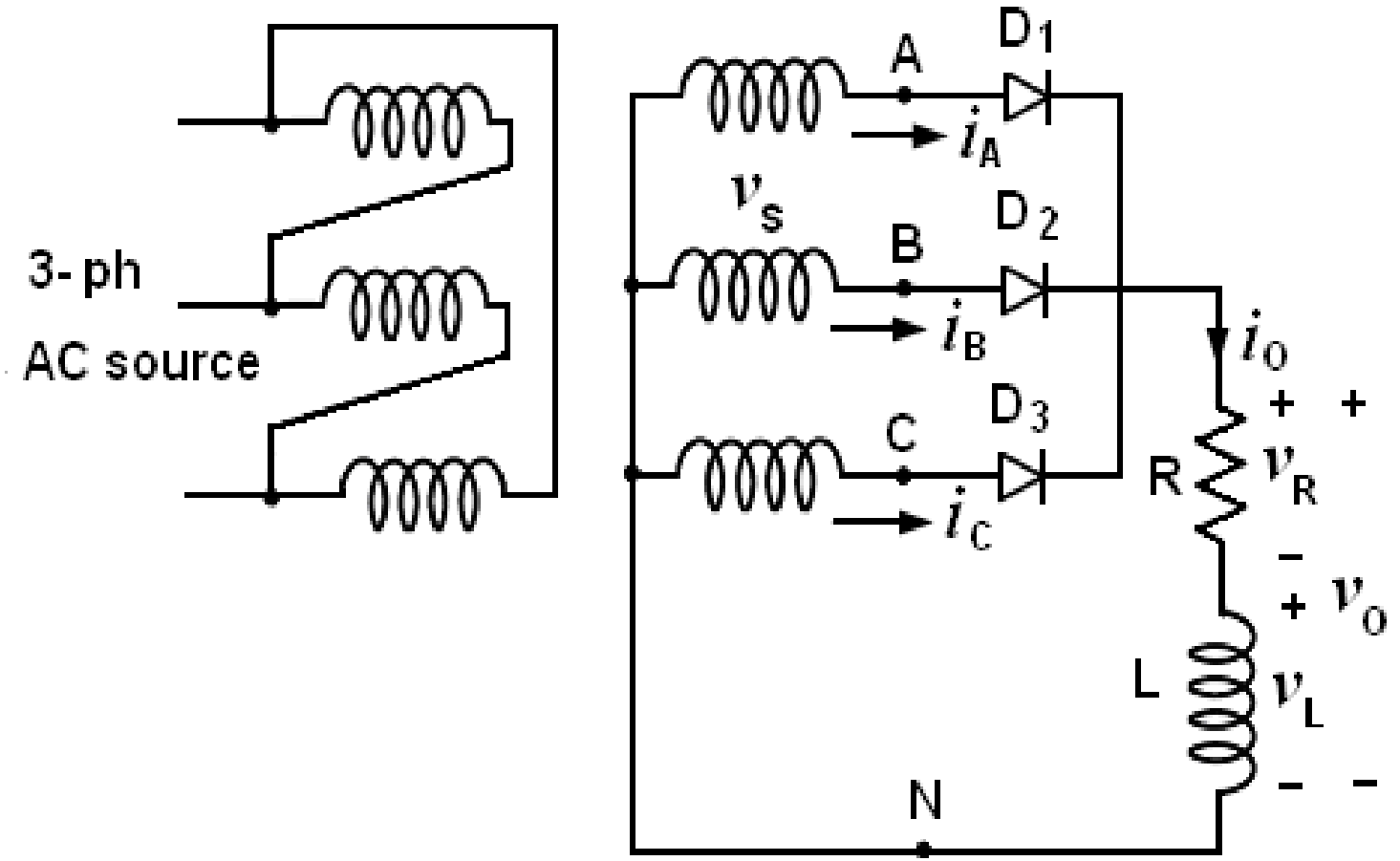

v_S
v_{AN}
v_{BN}
v_{CN}
$+V_m$
0
30°
90°
180°
270°
360°
450°
$-V_m$
v_{D2}
v_{AB}
v_{BC}
v_{CA}
v_{AB}
$+\sqrt{3}V_m$
v_{BA}
30°
90°
150°
270°
360°
510°
$-\sqrt{3}V_m$


v_{AB}
v_{BC}
v_{CA}
v_{AB}
v_{D_3}
$+\sqrt{3}\,V_m$
v_{CB}
30°
270°
390°
$-\sqrt{3}\,V_m$

This can limit its applications, particularly in cases where a higher DC voltage is required. The three-phase half-wave rectifier also produces high levels of harmonic distortion in the input current, which can cause problems such as increased losses in power transmission and distribution systems and increased electromagnetic interference.

The three-phase half-wave rectifier, while having a lower output voltage, offers benefits such as simplicity and lower cost due to its use of only three diodes. However, its high level of harmonic distortion in the input current can lead to issues like power losses in transmission and distribution systems and electromagnetic interference. Thus, when designing power electronics systems, it's essential to consider the trade-offs between the advantages and disadvantages of different rectifier types.

Three Phase Half-Wave Rectifier with RL Load



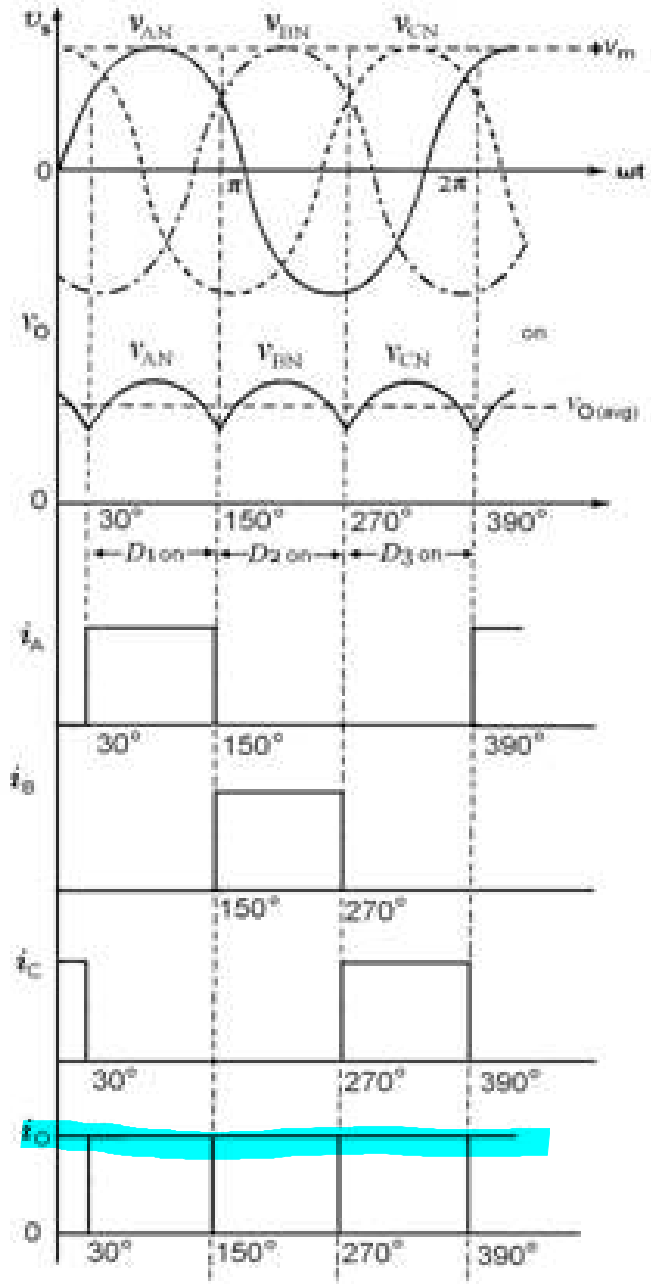
$\boldsymbol{For\ L \gg R}$

The average load voltage is:

$$V_{avg} = \frac{3\,V_{Pm}}{\pi}\,\sin\frac{\pi}{3}$$

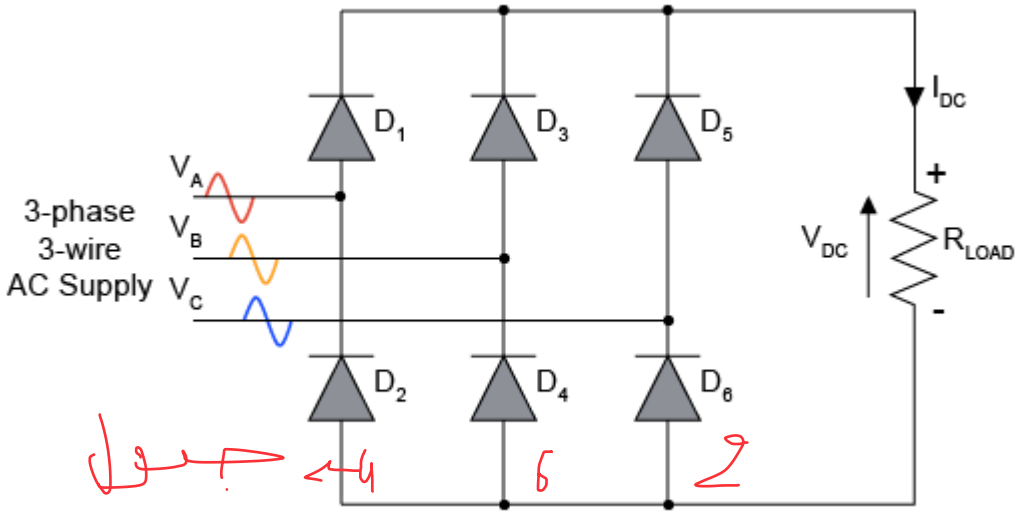
For a constant output current:

$$I_{avg} = I_{rms}$$


v_s
v_{AN}
v_{BN}
v_{CN}
$+V_m$
0
ωt
π
2π
v_O
on
v_{AN}
v_{BN}
v_{CN}
$V_{O(avg)}$
0
30°
150°
270°
390°
D_1 on
D_2 on
D_3 on
i_A
30°
150°
390°
i_B
150°
270°
i_C
30°
270°
390°
i_O
0
30°
150°
270°
390°

Three-Phase Full-Wave Rectifiers

The Figure bellow illustrates an uncontrolled three-phase diode rectifier circuit that utilizes a full-wave bridge configuration consisting of six diodes. The input of the circuit is linked to a three-phase AC power source, while the output is connected to a R load. The functioning of the circuit is described as follows.



1. Time period from 0° to 30°

   $V_C > VA > VB \Rightarrow$ Diodes D_5 and D_4 are ON

   Diode D_1 is OFF (Anode voltage V_A is lower than cathode voltage V_C)

   Diode D_2 is OFF (Anode voltage V_B is lower than cathode voltage V_A)

   Diode D_3 is OFF (Anode voltage V_B is lower than cathode voltage V_C)

   Diode D_6 is OFF (Anode voltage V_B is lower than cathode voltage V_C)

   So, the load voltage is equal to: $V_{load} = VC - VB$

2) Time period from 30° to 90°

$V_A > VC > VB \Rightarrow$ Diodes D_1 and D_4 are ON

Diode D_2 is OFF (Anode voltage V_B is lower than cathode voltage V_A)

Diode D_3 is OFF (Anode voltage V_B is lower than cathode voltage V_A)

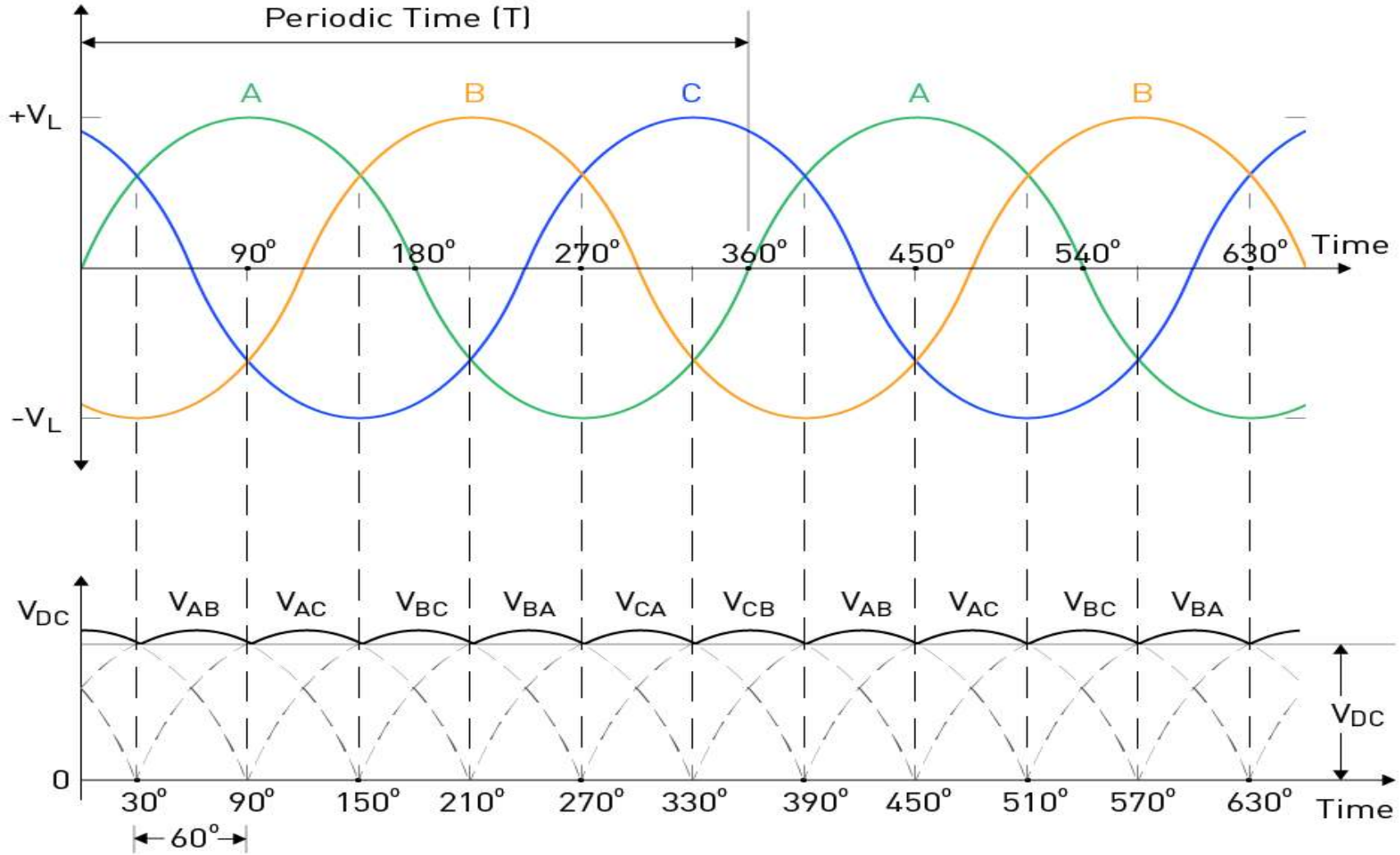
Diode D_5 is OFF (Anode voltage V_C is lower than cathode voltage V_A)

Diode D_6 is OFF (Anode voltage V_B is lower than cathode voltage V_C)

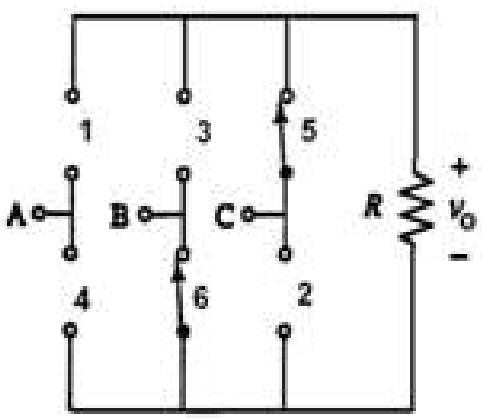
So, the load voltage is equal to: $V_{load} = VA - VB$

And so on… The waveform of the voltage at the output of the rectifier (on the load) is depicted in the figure above.

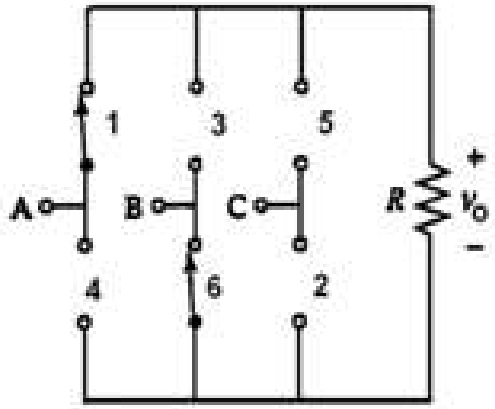
period	Highest Positive Voltage	Highest Negative Voltage	On Diodes	
			Odd-numbered	Even-numbered
0 – 60°	C	B	D_5	D_6
60 – 120°	A	B	D_1	D_6
120 – 180°	A	C	D_1	D_2
180 – 240°	B	C	D_3	D_2
240 – 300°	B	A	D_3	D_4
300 – 360°	C	A	D_5	D_4



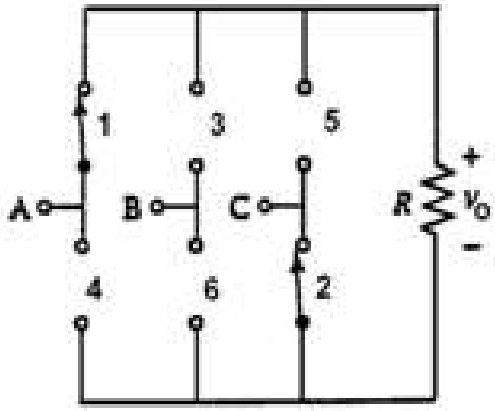
Output Voltage Waveform



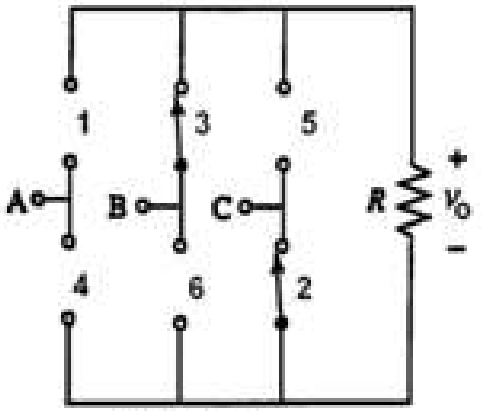
(a) $v_o = v_{CB}$



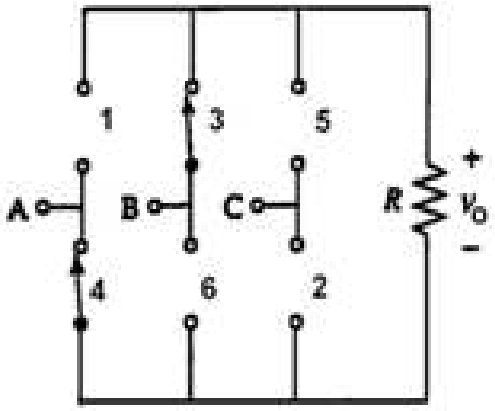
(b) $v_o = v_{AB}$



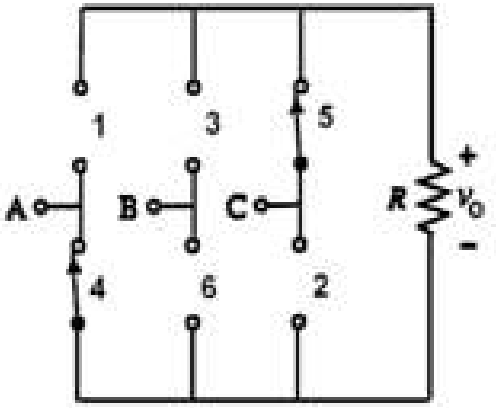
(c) $v_o = v_{AC}$



(d) $v_o = v_{BC}$



(e) $v_o = v_{BA}$



(f) $v_o = v_{CA}$

For star connection:

$$V_L = \sqrt{3}\ V_{Ph}$$

The average load voltage is:

$$V_{avg} = \frac{3}{\pi} \int_{\pi/3}^{2\pi/3} V_{Pm}\ \sin \omega t\ d\omega t = \frac{3\sqrt{3}}{\pi}\ V_{Pm} = \frac{3}{\pi} V_{Lm}$$

The rms output voltage is given by:

$$V_{rms} = \sqrt{\frac{6}{2\pi} \int_{\pi/3}^{2\pi/3} V_{Lm}^2 \sin^2 \omega t \, d\omega t} = V_L \sqrt{1 + \frac{3\sqrt{3}}{2\pi}} = V_{Pm} \sqrt{\frac{3}{2} + \frac{9\sqrt{3}}{4\pi}}$$

The load voltage form factor is:

$$FF = \frac{V_{rms}}{V_{avg}}$$

The ripple factor is:

$$\sqrt{(FF)^2 - 1}$$

The average diode current is:

$$I_{Davg} = \frac{I_{avg}}{3}$$

The rms diode current is:

$$I_{Drms} = \frac{I_{rms}}{\sqrt{3}}$$